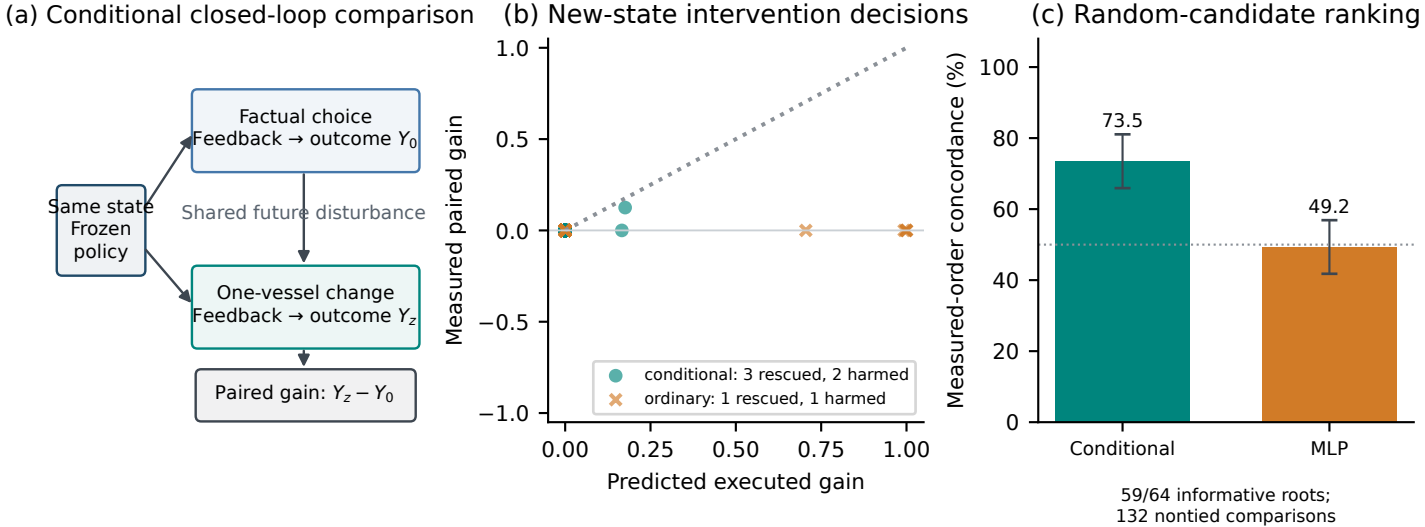

(a) Conditional closed-loop comparison
Factual choice
Feedback → outcome Y_0
Same state
Frozen
policy
Shared future disturbance
One-vessel change
Feedback → outcome Y_z
Paired gain: $Y_z - Y_0$
(b) New-state intervention decisions
Measured paired gain
1.0
0.5
0.0
−0.5
−1.0
0.00
0.25
0.50
0.75
1.00
Predicted executed gain
conditional: 3 rescued, 2 harmed
ordinary: 1 rescued, 1 harmed
(c) Random-candidate ranking
Measured-order concordance (%)
100
80
60
40
20
0
73.5
49.2
Conditional
MLP
59/64 informative roots;
132 nontied comparisons